

Security protocols in constrained environments

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TL;DR

System type	Such as	Will it work?	The issue
Low end embedded	Atmel 8-bit AVR (most Arduino), TI MSP-430	No	SRAM
Mid-high end embedded	Anything ARM based (e.g. STM Discovery, TI Stellaris) inc. Arduino Due	With some effort	Library, key and cipher suite wrangling
Linux OS	Raspberry Pi, BeagleBone, Arduino Yún	Yes	-

Agenda

- Anatomy of a security protocol
 - The key exchange dance
- Linux makes things easy
- Libraries for higher end microcontrollers
- SRAM on low end microcontrollers
- Summary

Which security protocols?

The 'S' protocols:

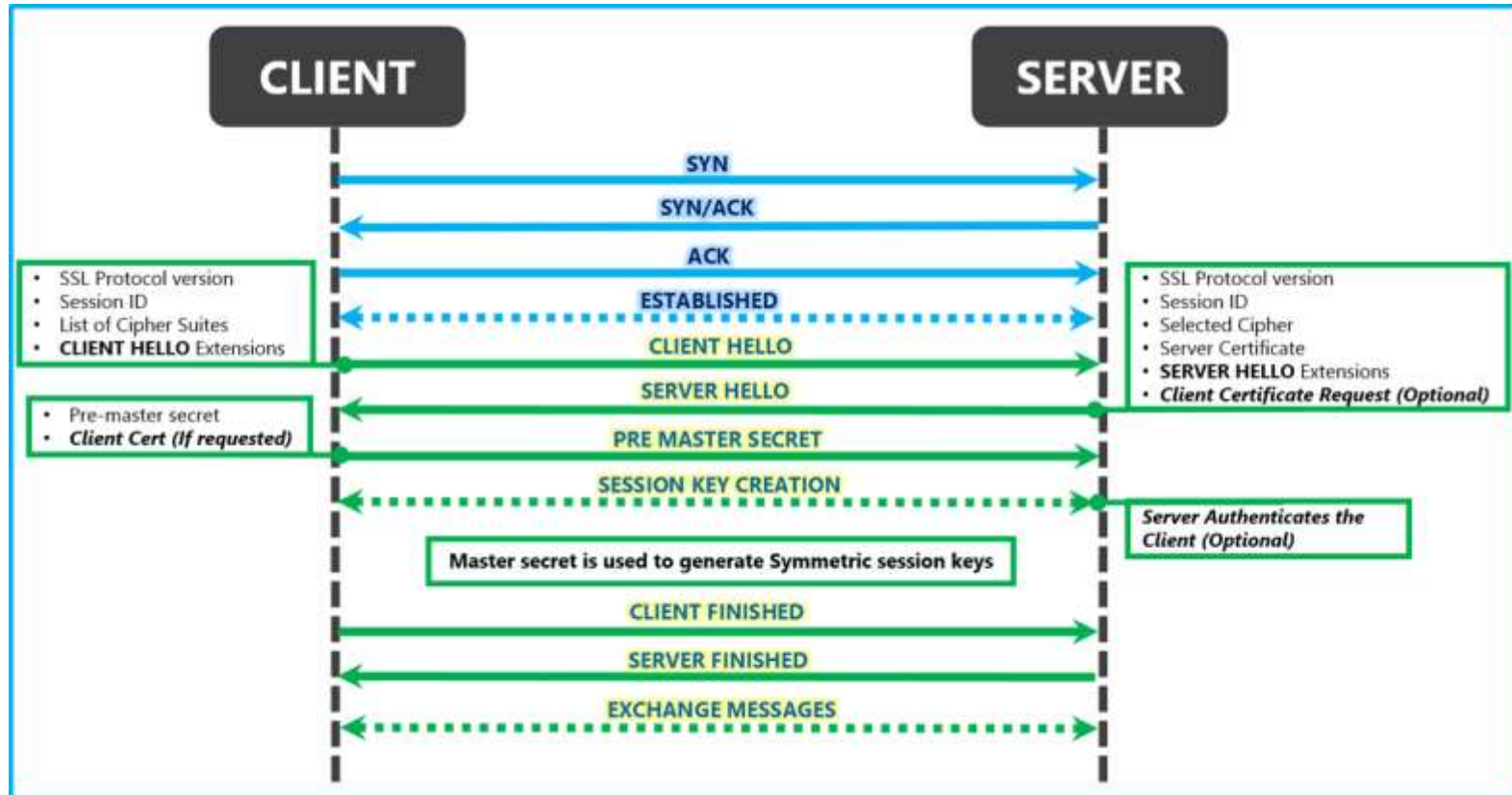
Secure Sockets Layer (SSL)

Superseded by Transport Layer **S**ecurity (TLS)

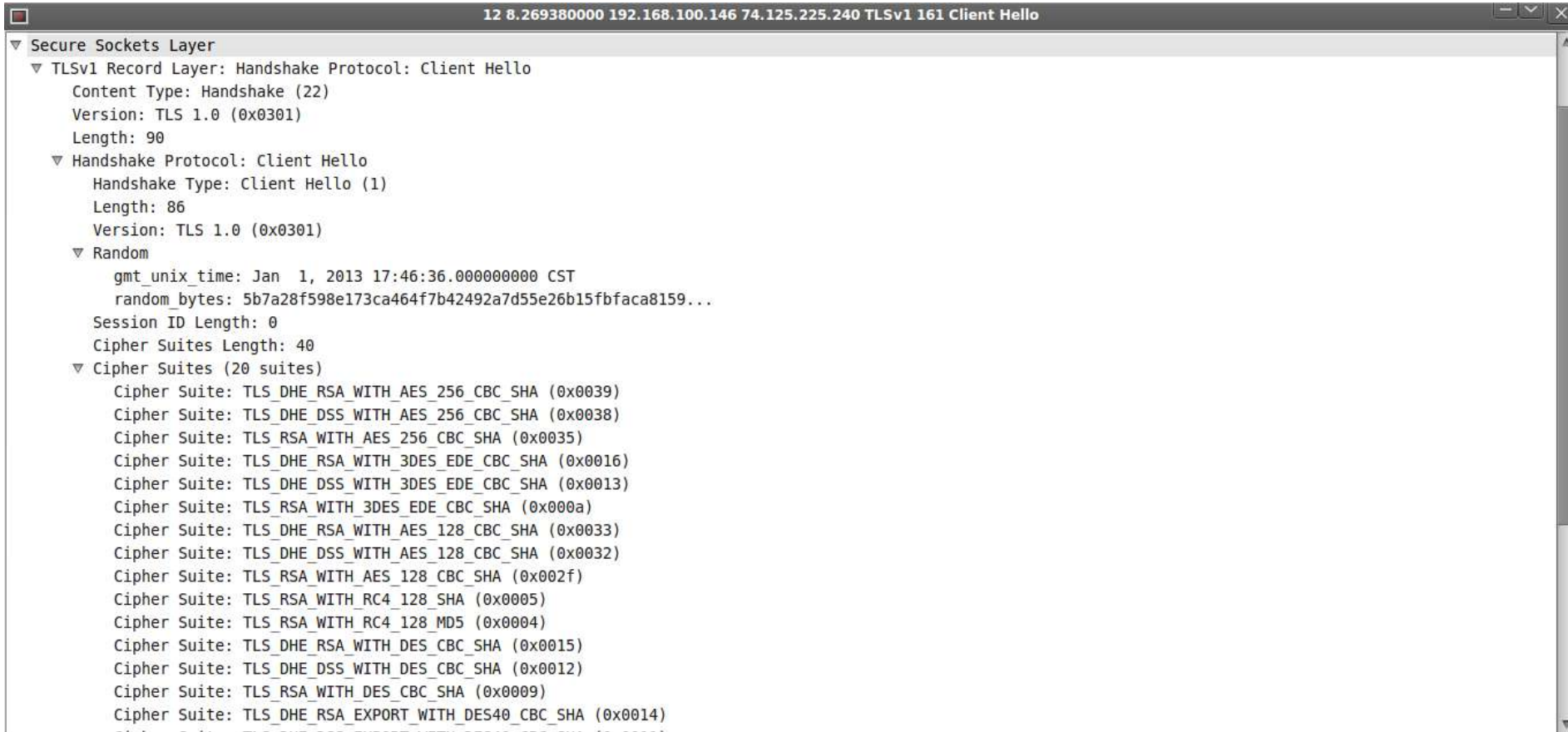
Secure SHell (SSH)

Internet Protocol Security (IPsec)

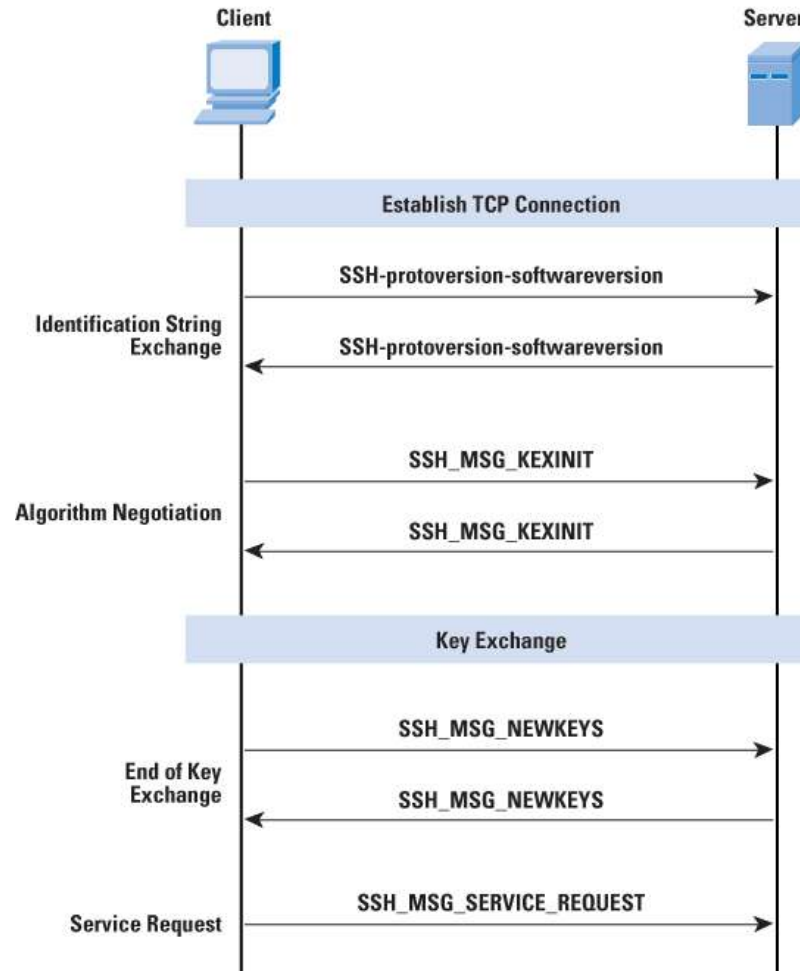
SSL Handshake



Client Hello



It's a similar story for SSH



and IPsec

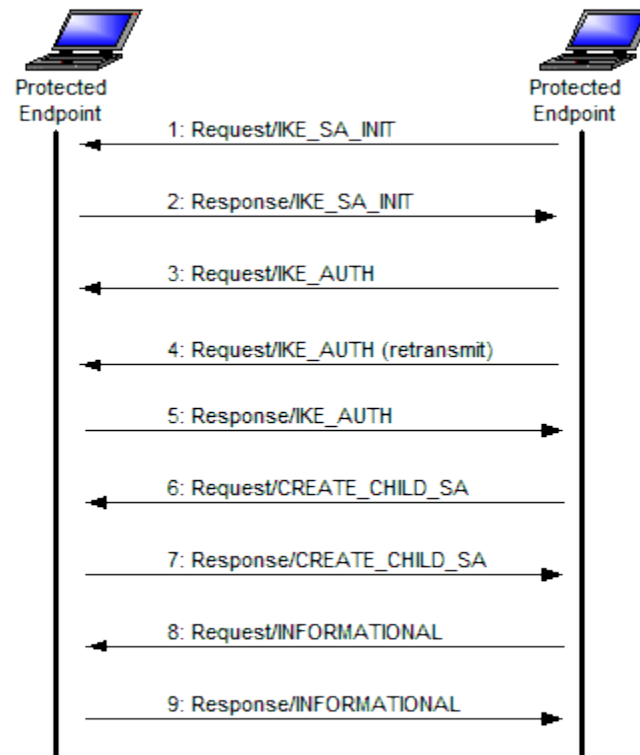


Figure 1: Sample IKE Exchange

Linux makes this easy

If not already built in to a particular distribution then use favourite package manager to get:



(no relation)

Things get trickier with embedded

But by no means impossible...



HIGHLIGHTS

- Up to **TLS 1.2** and **DTLS 1.2**
- Full client and server support
- Progressive list of supported ciphers
- Key and Certificate generation
- OCSP, CRL support
- Support Available

LIGHTWEIGHT

- Small Size: **20-100kB**
- Runtime Memory: **1-36kB**
- **20X** smaller than OpenSSL

PORTABLE

- Abstraction Layers (OS, Custom I/O, Standard C library, and [more](#))
- Simple API
- OpenSSL Compatibility Layer
- Long list of supported platforms



SPECIFICATIONS

- **< 50kB total footprint** with crypto provider
- SSL 3.0 and TLS 1.0, 1.1 and 1.2¹ server and client support
- Included crypto library - RSA, ECC,¹ 3DES, AES, ARC4, SHA1, MD5, RC2
- Assembly language optimizations for Intel, ARM and MIPS
- Cipher Suites - RC4-MD5, RC4-SHA, DES-CBC3-SHA, AES128-SHA, AES256-SHA
- Full support for session resumption/caching
- Session re-keying and cipher renegotiation
- Server and client X.509 certificate chain authentication
- Parsing of X.509 .pem and ASN.1 DER certificate formats
- PKCS#1.5, PKCS#5 PKCS#8 and PKCS#12 support for key formatting
- Certificate Revocation List (CRL) support
- SSH command line support¹
- Fully cross platform, portable codebase; [minimum use of system calls](#)
- Pluggable cipher suite interface
- Pluggable crypto provider interface
- Pluggable operating system and malloc interface
- TCP/IP optional
- Multithreading optional
- Only a handful of external APIs, all non-blocking
- Example client and server code included
- Clean, heavily commented code in portable C
- User and developer documentation

¹Included with commercial license



POLARSSL

Straightforward, Secure Communication

Easy integration with a small memory footprint

Tiny library. PolarSSL's memory footprint can get **as small as 30k** and averages below 110k.

Easy to understand and use with a clean API

We offer readable code and a logical and readable API with [API documentation](#) and [examples](#). Compare this to other SSL products around. You'll love it!

Easy to reduce and expand the code

PolarSSL has no global code and has minimal coupling between its modules. So it's easy to just grab part and drop it into an existing project or add new functionality.

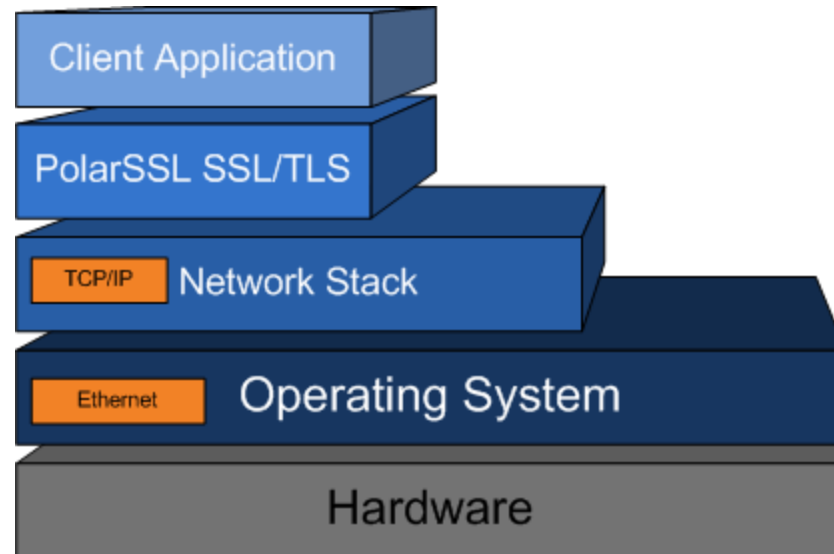
Easy to build with no external dependencies

Except for basic libc calls, PolarSSL has no external dependencies on other libraries.

Extremely portable

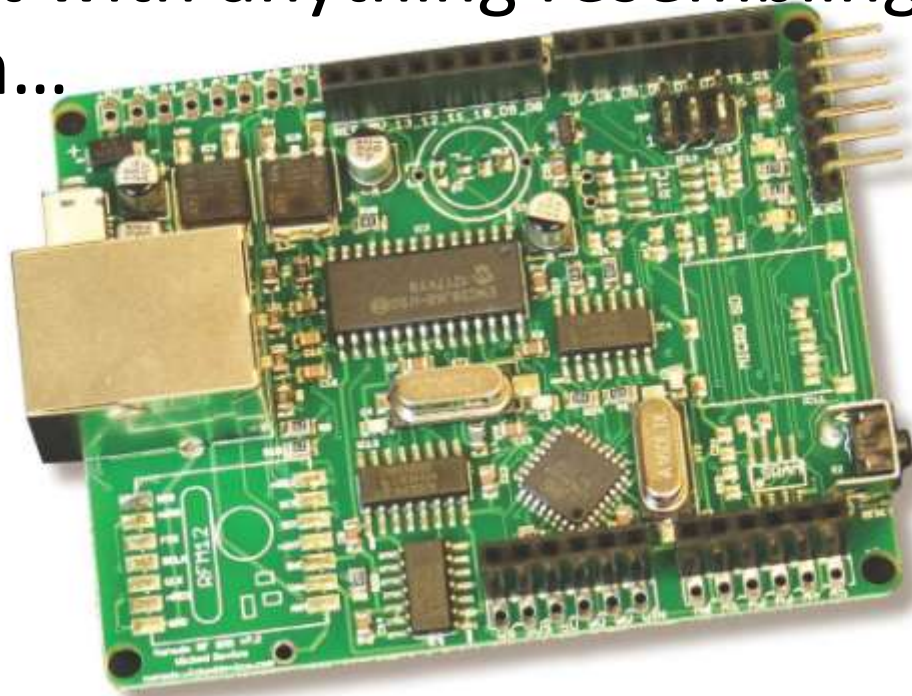
PolarSSL is used on many architectures, including x86/x64, ARM, PowerPC, MIPS and Motorola 68000 and OS'es like Windows, Linuxes, Unixes, Apple iOS, FreeRTOS, ThreadX, Android OS, Xbox, SEGGER embOS, OpenWRT, eCOS and much more.

Stack trades offs may be made



But those keys won't fit into 2K

At least not with anything resembling a useful application...



... Arduino struggles with MQTT and 1wire

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Questions?

Further reading

PolarSSL tutorial

<https://polarssl.org/kb/how-to/polarssl-tutorial>

AVR32753: AVR32 UC3 How to connect to an
SSL-server

<http://www.atmel.com/Images/doc32111.pdf>

STM32 Discovery: Porting Polar SSL

<http://hobbymc.blogspot.co.uk/2011/02/stm32-discovery-porting-polar-ssl.html>